

VRU TRADING BOT — ARCHITECTURE SPECIFICATION

Horizon Tech — VRU Trading Bot Research

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OVERVIEW

The VRU trading bot is built on the v23 register architecture as its core sequence model, retargeted from arithmetic reasoning to multi-class order flow classification.

This is NOT a price prediction system. It is a sequence classifier that detects institutional order flow patterns and outputs directional signals with confidence scores.

ARCHITECTURAL COMPONENTS

1. INPUT LAYER — Multi-Stream Feature Vector

Per timestep (1-minute bar resolution recommended):

Stream A — Price/Volume

- Normalized return: $(\text{close} - \text{open}) / \text{open}$
- Volume Z-score: $(\text{vol} - 30\text{d_mean_vol}) / 30\text{d_std_vol}$
- VWAP deviation
- High-low range normalized

Stream B — Options Flow

- Call/Put volume ratio (normalized by 30d average)
- Unusual sweep flag (binary: $\text{sweep} > 2 \text{ std above avg}$)
- Net options premium flow direction
- Put/Call open interest delta

Stream C — Dark Pool / Order Imbalance

- Dark pool print volume (normalized by float)
- Order book imbalance: $(\text{bid_vol} - \text{ask_vol}) / (\text{bid_vol} + \text{ask_vol})$
- Tape aggressor ratio: buyer-initiated vs seller-initiated

These 3 streams mirror the telemetry architecture from DPPU Document 6: Power (price) + Thermal lag (options) + Network (order flow) with coupling.

2. VRU CORE — v23 Register Architecture

Input projection: linear layer mapping feature vector to hidden dimension

Recurrent core: 6-slot register partitioned hidden state

- Slots 1-3: W_h (PHI-field) — memory/oscillatory boundary
- Slots 4-6: W_c (ALPHA-field) — carry/contraction
- Spectral clip: $RHO_H = 1/PHI$ for W_h , $RHO_C = PHI$ for W_c
- Additive phase injection between fields (NOT concatenation or gating)
- Per-slot register norm cap

State management: 4-tier buffer with circuit breaker

Target sequence length: 1,440 timesteps (1 trading day at 1-min bars)

Stress target: 5,000–15,000 timesteps for multi-day pattern detection

3. OUTPUT HEAD — Multi-Class Classifier

Linear projection from hidden state to 3 classes:

Class 0: No institutional flow — FLAT (no position)

Class 1: Accumulation signal — LONG

Class 2: Distribution signal — SHORT or EXIT LONG

Softmax activation → probability distribution over classes

Confidence score = $\max(\text{softmax output})$

Confidence thresholds:

< 0.60: No trade

0.60–0.75: Half position

> 0.75: Full position

4. POSITION SIZING MODULE

Kelly-adjacent sizing based on confidence score and recent win rate:

$\text{position_size} = \text{confidence_score} * \text{kelly_fraction} * \text{account_equity}$

kelly_fraction: start at 0.25 (quarter-Kelly for safety)

Hard limits:

- Max single position: 5% of account
- Max total exposure: 25% of account
- Max sector concentration: 10% of account

5. EXECUTION LAYER

Broker API integration (Alpaca for equities, CCXT for crypto)

Order type: limit orders only (no market orders — slippage control)

Execution latency target: <500ms from signal to order submission

Stop-loss: hard coded at -2% per position, no override by model

Take-profit: trailing stop at +1.5x ATR once +3% in profit

TRAINING PIPELINE

Stage 1 — Synthetic Order Flow Curriculum

Generate synthetic institutional execution patterns:

- Simulated VWAP execution over N bars with random N in [60, 1440]
- Inject into base price series with realistic noise
- Label: accumulation start bar, continuation bars, completion bar
- Train VRU to classify each timestep correctly
- Mastery gate: 90% accuracy on holdout before advancing

Stage 2 — Historical Data Training

Source: 2–5 years of 1-minute OHLCV + options flow + dark pool prints

Label generation: identify confirmed institutional moves via 13F delta confirmation

Loss function: weighted cross-entropy (Class 0 heavily weighted to prevent overtrading)

Validation: walk-forward validation only (no random train/test split — look-ahead bias)

Stage 3 — Paper Trading

Live data, simulated execution

Minimum 90 days before any real capital

Track: signal accuracy, Sharpe, max drawdown, win rate by confidence bucket

Stage 4 — Live Trading

Start: minimum viable capital (\$5,000–\$25,000)

Scale only after 6 months live track record

REGRESSION DETECTION

Inherited from v23 curriculum harness:

- All prior stage accuracy monitored continuously
- Circuit breaker halts trading if any stage drops below mastery threshold
- JSONL audit trail on all signals, positions, and outcomes

This is the same mechanism as v23's regression detection on prior stages, applied to live trading performance rather than training tasks.